COURSE INFORMATION

School/Faculty:	Computing	**Page:**	1 of 5
Program name:	Bachelor in Software Engineering		
Course code:	SCSE 1013	**Academic Session/Semester:**	2024/25/1
Course name:	Fundamental Programming Concepts	**Pre/co requisite (course name and code, if applicable):**	-
Credit hours:	3		

Course synopsis	This course covers problem solving techniques and the fundamentals of programming. Students will learn to apply programming concepts towards solving problems. Then, students will learn of the techniques of problem solving in programming before implementing the techniques. Furthermore, students will learn the basics of programming and control structures involved. Moreover, students will be introduced to functions, arrays and input and output files. Students will also learn about pointers and structured data, which includes structured and enumerated data types. Students are required to develop programs using computer programming language, in order to solve simple to moderate problems. At the end of this course, students are expected to mastery in problem-solving skill and programming skill.			
Course coordinator (if applicable)	Dr. Sim Hiew Moi			
Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail (@utm.my)**
	Dr Nor Azizah Binti Sa'adon			
	Dr. Noraini Ibrahim			
	Dr. Muhammad Khatibsyarbini			

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO	PLO (Code)	Taxonomies And generic skills	T&L methods	Assessment methods
CLO1	Apply programming concepts using tools to solve problems in a given time frame	PLO3 (PS)	C3	Lecture, Active Learning	L, T, Q, F
CLO2	Communicate project solutions through project demo and presentation.	PLO5 (CS)	CS5	Lecture, Active Learning, Project-based learning	A
CLO3	Use appropriate programming tools to develop a simple to complex program to solve a problem.	PLO6 (DS)	P4	Active Learning, Project-based learning	L, A
*** T - Test; L - Lab Exercise; F - Final Exam; A - Assignment					

Prepared by: Name: Dr. Sim Hiew Moi Signature: Date: 5/10/2024	Certified by: Name: Radziah Mohamad Signature: Date: 26/6/2022

Details on Innovative T&L practices:

No.	Type	Implementation
1.	Active learning - Live Coding Demonstration - Code Review using Online Coding Platform - Case Studies	Conducted through in-class activities Live Coding Demonstration – The instructor codes live, explaining the thought process, problem-solving strategies, and best practices. Students can ask questions in realtime. Code Review using Online Coding Platform – Students introduced with Github platform to do code review. Case Studies - Present real-world programming scenarios or challenges as case studies. Discuss and analyze different approaches and solutions.
2.	Project-based learning	Conducted through assignments. Students in a group of 2 are given 3 projects that required programming solutions. Student will be a given a similar guided project throughout Assignment 1 until Assignment 3. Assignment 1 will be focusing on assessing student's problem solving skill. Meanwhile, Assignment 2 & 3 will be focusing on assessing student's psychomotor skill in programming.

Weekly Schedule:

Week 1	**CHAPTER 1. COMPUTATIONAL THINKING AND PROBLEM-SOLVING** Computational thinking, logical thinking, algorithm thinking, Define problem, Decomposition, patterns and generalisation
Week 2	**CHAPTER 2. FUNDAMENTAL PROGRAMMING CONCEPTS** Problem-Solving Process, Programming paradigms, Programming Process, Procedural and Object-Oriented Programming, Pseudo codes, Flowcharts
Week 3	**CHAPTER 2. FUNDAMENTAL PROGRAMMING CONCEPTS** Algorithm and Flowchart
Week 4	**CHAPTER 3. BASIC OF PROGRAMMINGG** Data Types, Operators, Input and Output
Week 5	**CHAPTER 4. CONTROL STRUCTURES** Selection Control Structures and Repetition Control Structures
Week 6	**CHAPTER 5. FUNCTION** Predefined Function
Week 7	**CHAPTER 5. FUNCTION** User-Defined Function
Week 8	**MID SEMESTER BREAK**

Week 9 (Test Chapter 1 – Chapter 5) 3/12/2024 (5pm to 7pm)	**CHAPTER 6. ARRAY** Basic of Array, Passing and Returning Arrays to/ from Functions
Week 10	**CHAPTER 7. INPUT AND OUTPUT FILE**
Week 11	**CHAPTER 8. POINTERS** Basic of Pointers, Array and Pointers
Week 12	**CHAPTER 8. POINTERS** Function and Pointers
Week 13	**CHAPTER 9. STRUCTURED DATA** Structured and Enumerated Data Types
Week 14 - 15	**CHAPTER 10. TRANSFORMATION FROM STRUCTURED TO OBJECT-ORIENTED PARADIGM** **Object-Oriented Programming Principles** **- Class and Object Manipulations** **- Association/Aggregation/Composition** **- Inheritance and Polymorphisms**

Transferable skills (generic skills learned in course of study which can be useful and utilised in other settings):

Computational Thinking skills
Problem-solving skills
Programming skills

Student learning time (SLT) details:

Distribution of student Learning Time (SLT) Course content outline	Guided Learning (Face to Face)				Teaching and Learning Activities: Guided Learning Non-Face to Face	Teaching and Learning Activities: Independent Learning Non-Face to face	TOTAL SLT
CLO	**L**	**T**	**P**	**O**			
CLO 1	28h	7h			20h	30h	**85h**
CLO 2				3h	15h	3h	**21h**
CLO 3		2h	2h		2h	3h	**9h**
Total SLT	**28h**	**9h**	**2h**	**3h**	**37h**	**36h**	**115h**

	Continuous Assessment	PLO	Percentage	Total SLT
1	Lab Exercise (2) - Concepts	PLO3 (PS)	10	As in CLO 1 (1h)
2	Lab Exercise (2) - Tools	PLO6 (DS)	10	As in CLO 3 (3h)
3	Quizzes (2)	PLO3 (PS)	10	As in CLO 1 (2h30m)
3	Test	PLO3 (PS) PLO6 (DS)	20	**2h**
4	Assignment 1 & 2	PLO3 (PS)	20	As in CLO 1 (5h) As in CLO2 (2h) As in CLO 3 (3h)

Final Assessment			Percentage	Total SLT
1	Final Examination	PLO3 (PS)	30	**3h**
Grand Total SLT				**120h**

L: Lecture, T: Tutorial, P: Practical, O: Others

** To pass this course, students must pass both Continuous and Final assessment*

Special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room):

Computer Programming Lab and Software: C++ Compiler

Learning resources:

Text book (if applicable)

Tony Gaddis (2020), Starting out with C++: Early Objects, Brief Version, 10th edition. Pearson Education.
D. S. Malik (2014), C++ Programming: From Problem Analysis to Program Design, 7th edition. Cengage Learning.

Additional references

Walter Savitch (2015), Problem Solving with C++. 9th edition. Pearson Education.
H.M. Deitel and P.J. Deitel (2014), C++ How to Program. 9th edition. Pearson Education.
https://www.amazon.com/Computational-Thinking-beginners-problem-solving-programming-ebook/dp/B072MGKS96

Online

http://elearning.utm.my

Academic honesty and plagiarism: *(Below is just a sample)*

Lab exercises and assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES). Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of **zero** for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (Course policy, any specific instruction etc.):

1. Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.
2. Students are required to behave and follow the University's dressing regulation and etiquette all the time.
3. Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.

4. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted three days after the due date will not be accepted.
5. Make up exam will not be given, except to students who are sick and submit medical certificate which is confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.

Disclaimer:

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ONLINE LEARNING & TEACHING (L&T) PLAN

School/Faculty:	**Faculty of Computing**	**Page:**	
Program name:	Bachelor in Software Engineering with Honours		
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REVIEW OF L&T ACTIVITIES TO INCLUDE ONLINE LEARNING						
Course learning outcome	**Guided Learning FTF hours (from CI)**	**Guided Learning FTF hours completed**	**Online Learning hours**			
			Activities	**Type of time spent**	**Estimated time**	**Total time**
CLO1 Apply programming concepts using tools to solve problems in a given time frame.	*35h*	*27h*	*8h*			
			Live lecture (recorded lecture provided)	*The time required to consume content*	*6 hours*	*6 hours*
			Live Interaction with students	*The time spent in synchronous live interaction*	*1 hour*	*1 hour*
			Live tutorial	*Time spent in synchronous live tutorial*	*1 hours*	*1 hours*
CLO2 Communicate project solutions through project demo and presentation.	*3h*	*0*	*0h*			
CLO3 Use appropriate programming tools to develop a simple to complex program to solve a problem.	*4h*	*0*	*0h*			

REVIEW OF ASSESSMENT PLAN				
Before (from CI)		**Revised***		
Continuous Assessment			Percentage	Total SLT
1	Lab Exercise 1	*Lab Exercise 1*	5	As in CLO1
2	Lab Exercise 2	*Lab Exercise 2*	5	As in CLO2
3	Test 1	*Test*	20	As in CLO1 & CLO2
4	Assignment 1 - 3	*Assignment 1 - 3*	30	As in CLO2
Summative Assessment			Percentage	
1	Final Examination	*Final Examination*	30	**3h**

** All CLO should be maintained but changes can be made on the type of assessment. If changes have been made on the assessment type which affects the SLT, please adjust the SLT in the CI accordingly.*

Prepared by: Name: Sim Hiew Moi Signature: Date: 13/5/2024	Certified by: Name: Radziah Mohamad Signature: Date: 13/5/2024